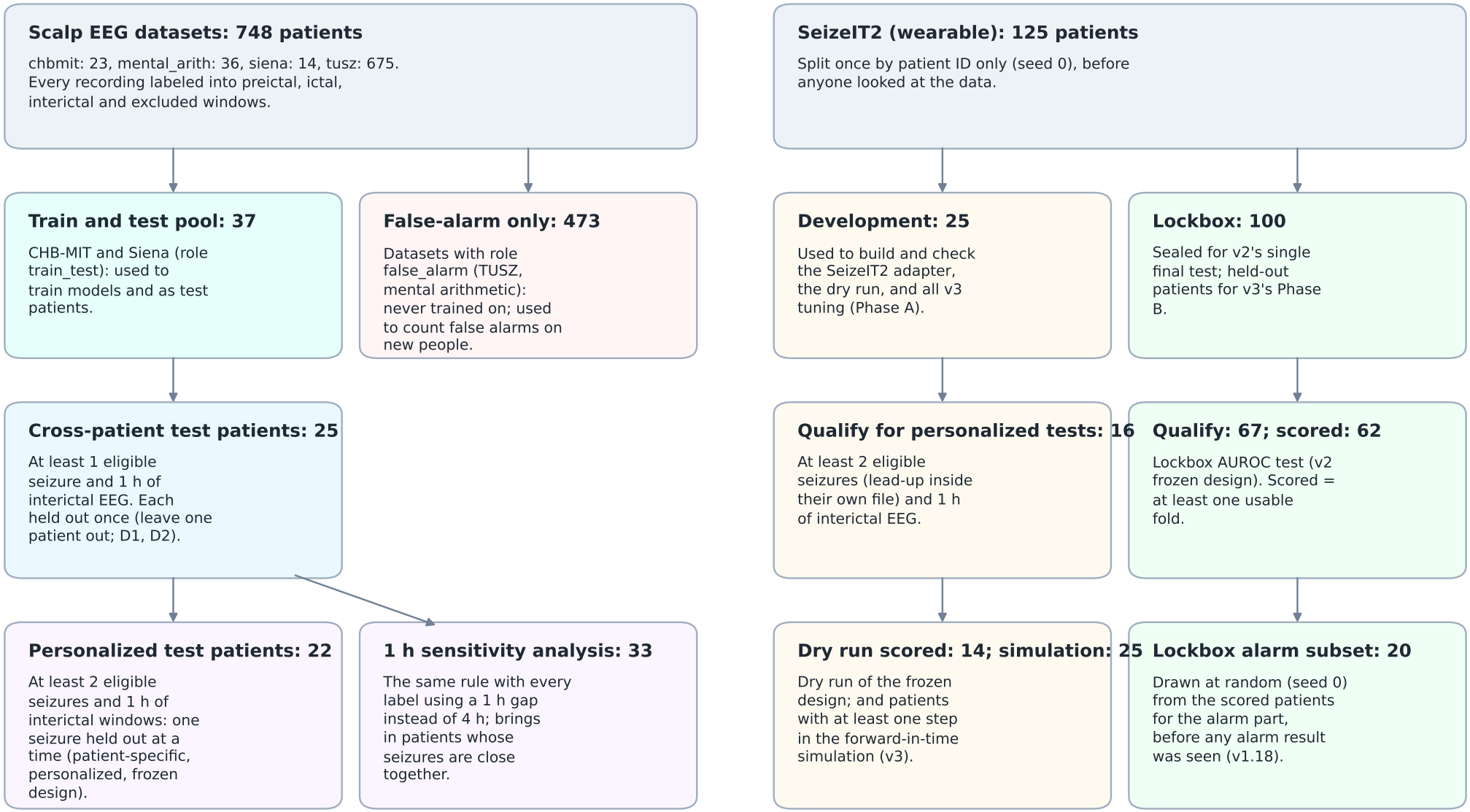
How patients were categorized

Counts are read from the project's result files. Boxes on the left: scalp EEG (D1, D2). Boxes on the right: SeizeIT2 wearable EEG (lockbox, v3).



Why patients drop out: a seizure is eligible only if at least 90% of its 30 min lead-up was recorded (SeizeIT2: inside its own file), and seizures less than 30 min apart count as one event. Interictal EEG must be at least 4 h from any seizure. Patients without enough of either can't be tested in that way.

The rules behind each category

Category	Rule
Preictal window	30 min to 5 s before a seizure onset (the warning period).
Ictal window	During a seizure.
Interictal window	At least 4 h from any seizure (normal EEG).
Excluded window	Everything else (too close to a seizure to be normal, not in a warning period).
Event	Seizures less than 30 min apart are grouped; the first one's onset counts.
Eligible event	At least 90% of its 30 min lead-up is recorded (SeizeIT2: inside the seizure's own file).
Role train_test	CHB-MIT and Siena patients: can be trained on and tested.
False-alarm only	TUSZ patients (and the mental arithmetic set): no eligible seizures; used only to measure false alarms on people the model never saw.
Cross-patient test patient	Role train_test, at least 1 eligible event and 1 h of interictal EEG. Tested by leaving that patient out (D1, D2).
Personalized test patient	At least 2 eligible events and 1 h of interictal windows. Tested by leaving one seizure out at a time, with buffers (D2, frozen design).
SeizeIT2 development	25 patients drawn by ID only (seed 0): adapter, dry run, all v3 tuning.
SeizeIT2 lockbox	The other 100: v2's single final test; v3's held-out Phase B patients.
Qualifies (SeizeIT2)	At least 2 eligible events and 1 h of interictal EEG.
Scored	Qualified and had at least one fold with usable training data.
Simulation patient (v3)	Has at least one forward-in-time step (calibration, or a retraining before the next seizure).

CHB-MIT and Siena, patient by patient (1)

Green rows were test patients. 'Training only' patients only contributed training data (too few eligible seizures or too little interictal EEG to be tested).

Patient	Dataset	Role	Hours	Seizures	Events	Eligible	Interictal h	Tested in
chbmit:chb01	chbmit	train_test	73.4	11	11	10	37.8	✓ cross-patient, personalized
chbmit:chb02	chbmit	train_test	35.3	3	3	2	27.0	✓ cross-patient, personalized
chbmit:chb03	chbmit	train_test	38.0	7	7	6	27.4	✓ cross-patient, personalized
chbmit:chb04	chbmit	train_test	156.1	4	4	4	131.9	✓ cross-patient, personalized
chbmit:chb05	chbmit	train_test	39.0	5	5	5	14.4	✓ cross-patient, personalized
chbmit:chb06	chbmit	train_test	66.7	10	10	10	24.8	✓ cross-patient, personalized
chbmit:chb07	chbmit	train_test	67.1	3	3	3	51.3	✓ cross-patient, personalized
chbmit:chb08	chbmit	train_test	20.0	5	5	5	1.0	✓ cross-patient, 1 h analysis only
chbmit:chb09	chbmit	train_test	67.9	4	4	4	48.6	✓ cross-patient, personalized
chbmit:chb10	chbmit	train_test	50.0	7	7	6	24.3	✓ cross-patient, personalized
chbmit:chb11	chbmit	train_test	34.8	3	3	1	32.0	✓ cross-patient
chbmit:chb12	chbmit	train_test	23.7	40	11	8	0.0	✓ 1 h analysis only
chbmit:chb13	chbmit	train_test	33.0	12	8	5	15.1	✓ cross-patient, personalized
chbmit:chb14	chbmit	train_test	26.0	8	6	6	4.7	✓ cross-patient, personalized
chbmit:chb15	chbmit	train_test	40.0	20	14	9	1.6	✓ cross-patient, personalized
chbmit:chb16	chbmit	train_test	19.0	10	5	5	5.6	✓ cross-patient, personalized
chbmit:chb17	chbmit	train_test	21.0	3	3	3	12.8	✓ cross-patient, personalized
chbmit:chb18	chbmit	train_test	35.6	6	5	4	26.4	✓ cross-patient, personalized
chbmit:chb19	chbmit	train_test	29.9	3	3	2	27.0	✓ cross-patient, personalized
chbmit:chb20	chbmit	train_test	27.6	8	6	5	19.1	✓ cross-patient, personalized
chbmit:chb22	chbmit	train_test	31.0	3	3	2	17.1	✓ cross-patient, personalized
chbmit:chb23	chbmit	train_test	26.6	7	5	5	14.2	✓ cross-patient, personalized
chbmit:chb24	chbmit	train_test	21.3	16	13	12	0.0	✓ 1 h analysis only
siena:PN00	siena	train_test	3.2	5	5	0	0.0	training only
siena:PN01	siena	train_test	13.5	2	2	2	2.0	✓ cross-patient, personalized
siena:PN03	siena	train_test	24.2	2	2	2	12.4	✓ cross-patient, personalized
siena:PN05	siena	train_test	6.0	3	3	3	0.0	✓ 1 h analysis only
siena:PN06	siena	train_test	12.1	5	5	5	0.0	✓ 1 h analysis only
siena:PN07	siena	train_test	8.7	1	1	1	2.1	✓ cross-patient
siena:PN09	siena	train_test	6.8	3	3	3	0.0	✓ 1 h analysis only
siena:PN10	siena	train_test	18.7	10	10	10	0.0	✓ 1 h analysis only
siena:PN11	siena	train_test	2.4	1	1	1	0.0	training only
siena:PN12	siena	train_test	6.1	4	4	2	0.0	✓ 1 h analysis only
siena:PN13	siena	train_test	8.7	3	3	3	0.0	✓ 1 h analysis only

CHB-MIT and Siena, patient by patient (2)

Green rows were test patients. 'Training only' patients only contributed training data (too few eligible seizures or too little interictal EEG to be tested).

Patient	Dataset	Role	Hours	Seizures	Events	Eligible	Interictal h	Tested in
siena:PN14	siena	train_test	20.5	4	4	4	5.8	✓ cross-patient, personalized
siena:PN16	siena	train_test	4.9	2	2	2	0.0	✓ 1 h analysis only
siena:PN17	siena	train_test	5.1	2	2	2	0.0	✓ 1 h analysis only

TUSZ, summary

Patients	Role	Hours	Seizures	Eligible events	Used for
675	false_alarm	1475	2663	0	✓ false-alarm counting only (never trained on)

SeizeIT2, patient by patient (1)

Seizure counts and qualification come from the development feasibility report; lockbox patients' details weren't tabulated per patient (shown as –).

Patient	Group	EEG h	Seizures	Eligible	Interictal h	Qualifies	Tested in
seizeit2:sub-001	development	106.6	4	4	74.4	yes	✓ dry run, v3 simulation
seizeit2:sub-002	development	99.2	15	14	47.9	yes	✓ dry run, v3 simulation
seizeit2:sub-003	lockbox	–	–	–	–	–	not tested
seizeit2:sub-004	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-005	development	121.2	2	2	105.0	yes	✓ dry run, v3 simulation
seizeit2:sub-006	lockbox	–	–	–	–	–	not tested
seizeit2:sub-007	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-008	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-009	development	99.4	1	1	92.8	no	✓ v3 simulation
seizeit2:sub-010	lockbox	–	–	–	–	–	not tested
seizeit2:sub-011	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-012	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-013	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-014	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-015	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-016	lockbox	–	–	–	–	–	not tested
seizeit2:sub-017	lockbox	–	–	–	–	–	not tested
seizeit2:sub-018	lockbox	–	–	–	–	–	not tested
seizeit2:sub-019	lockbox	–	–	–	–	–	not tested
seizeit2:sub-020	development	115.7	1	1	107.6	no	✓ v3 simulation
seizeit2:sub-021	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-022	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-023	lockbox	–	–	–	–	–	not tested
seizeit2:sub-024	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-025	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-026	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-027	lockbox	–	–	–	–	–	not tested
seizeit2:sub-028	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-029	development	116.9	4	4	90.8	yes	✓ dry run, v3 simulation
seizeit2:sub-030	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-031	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-032	lockbox	–	–	–	–	–	not tested
seizeit2:sub-033	development	113.3	1	1	105.2	no	✓ v3 simulation
seizeit2:sub-034	development	95.3	30	23	37.4	yes	✓ dry run, v3 simulation

SeizeIT2, patient by patient (2)

Seizure counts and qualification come from the development feasibility report; lockbox patients' details weren't tabulated per patient (shown as –).

Patient	Group	EEG h	Seizures	Eligible	Interictal h	Qualifies	Tested in
seizeit2:sub-035	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-036	lockbox	–	–	–	–	–	not tested
seizeit2:sub-037	lockbox	–	–	–	–	–	not tested
seizeit2:sub-038	lockbox	–	–	–	–	–	not tested
seizeit2:sub-039	lockbox	–	–	–	–	–	not tested
seizeit2:sub-040	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-041	lockbox	–	–	–	–	–	not tested
seizeit2:sub-042	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-043	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-044	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-045	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-046	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-047	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-048	lockbox	–	–	–	–	–	not tested
seizeit2:sub-049	lockbox	–	–	–	–	–	not tested
seizeit2:sub-050	development	70.1	3	2	59.4	yes	✓ v3 simulation
seizeit2:sub-051	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-052	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-053	development	25.8	2	2	13.3	yes	✓ dry run, v3 simulation
seizeit2:sub-054	lockbox	–	–	–	–	–	not tested
seizeit2:sub-055	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-056	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-057	development	61.2	1	1	53.7	no	✓ v3 simulation
seizeit2:sub-058	lockbox	–	–	–	–	–	not tested
seizeit2:sub-059	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-060	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-061	lockbox	–	–	–	–	–	not tested
seizeit2:sub-062	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-063	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-064	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-065	development	88.2	2	2	72.0	yes	✓ dry run, v3 simulation
seizeit2:sub-066	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-067	development	35.6	4	3	27.5	yes	✓ v3 simulation
seizeit2:sub-068	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms

SeizeIT2, patient by patient (3)

Seizure counts and qualification come from the development feasibility report; lockbox patients' details weren't tabulated per patient (shown as –).

Patient	Group	EEG h	Seizures	Eligible	Interictal h	Qualifies	Tested in
seizeit2:sub-069	lockbox	–	–	–	–	–	not tested
seizeit2:sub-070	development	141.7	2	1	126.7	no	✓ v3 simulation
seizeit2:sub-071	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-072	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-073	development	280.7	75	47	126.3	yes	✓ dry run, v3 simulation
seizeit2:sub-074	development	154.7	9	9	115.5	yes	✓ dry run, v3 simulation
seizeit2:sub-075	lockbox	–	–	–	–	–	not tested
seizeit2:sub-076	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-077	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-078	lockbox	–	–	–	–	–	not tested
seizeit2:sub-079	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-080	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-081	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-082	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-083	development	63.3	1	1	54.8	no	✓ v3 simulation
seizeit2:sub-084	lockbox	–	–	–	–	–	not tested
seizeit2:sub-085	development	55.9	3	1	40.7	no	✓ v3 simulation
seizeit2:sub-086	development	89.2	4	3	58.8	yes	✓ dry run, v3 simulation
seizeit2:sub-087	lockbox	–	–	–	–	–	not tested
seizeit2:sub-088	lockbox	–	–	–	–	–	not tested
seizeit2:sub-089	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-090	development	3.5	4	3	0.0	no	✓ v3 simulation
seizeit2:sub-091	lockbox	–	–	–	–	–	not tested
seizeit2:sub-092	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-093	lockbox	–	–	–	–	–	not tested
seizeit2:sub-094	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-095	lockbox	–	–	–	–	–	not tested
seizeit2:sub-096	lockbox	–	–	–	–	–	not tested
seizeit2:sub-097	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-098	lockbox	–	–	–	–	–	not tested
seizeit2:sub-099	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-100	development	112.3	2	2	96.2	yes	✓ dry run, v3 simulation
seizeit2:sub-101	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-102	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms

SeizeIT2, patient by patient (4)

Seizure counts and qualification come from the development feasibility report; lockbox patients' details weren't tabulated per patient (shown as –).

Patient	Group	EEG h	Seizures	Eligible	Interictal h	Qualifies	Tested in
seizeit2:sub-103	development	89.6	15	10	29.6	yes	✓ dry run, v3 simulation
seizeit2:sub-104	lockbox	–	–	–	–	–	not tested
seizeit2:sub-105	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-106	lockbox	–	–	–	–	–	not tested
seizeit2:sub-107	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-108	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-109	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-110	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-111	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-112	development	66.7	7	7	35.6	yes	✓ dry run, v3 simulation
seizeit2:sub-113	development	95.2	10	9	40.3	yes	✓ dry run, v3 simulation
seizeit2:sub-114	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-115	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-116	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-117	lockbox	–	–	–	–	–	✓ lockbox AUROC
seizeit2:sub-118	development	124.6	2	1	108.4	no	✓ v3 simulation
seizeit2:sub-119	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-120	lockbox	–	–	–	–	–	not tested
seizeit2:sub-121	lockbox	–	–	–	–	–	not tested
seizeit2:sub-122	lockbox	–	–	–	–	–	✓ lockbox AUROC, lockbox alarms
seizeit2:sub-123	lockbox	–	–	–	–	–	not tested
seizeit2:sub-124	lockbox	–	–	–	–	–	not tested
seizeit2:sub-125	lockbox	–	–	–	–	–	not tested